

report

Link to the source code: [GitHub Repository](#)

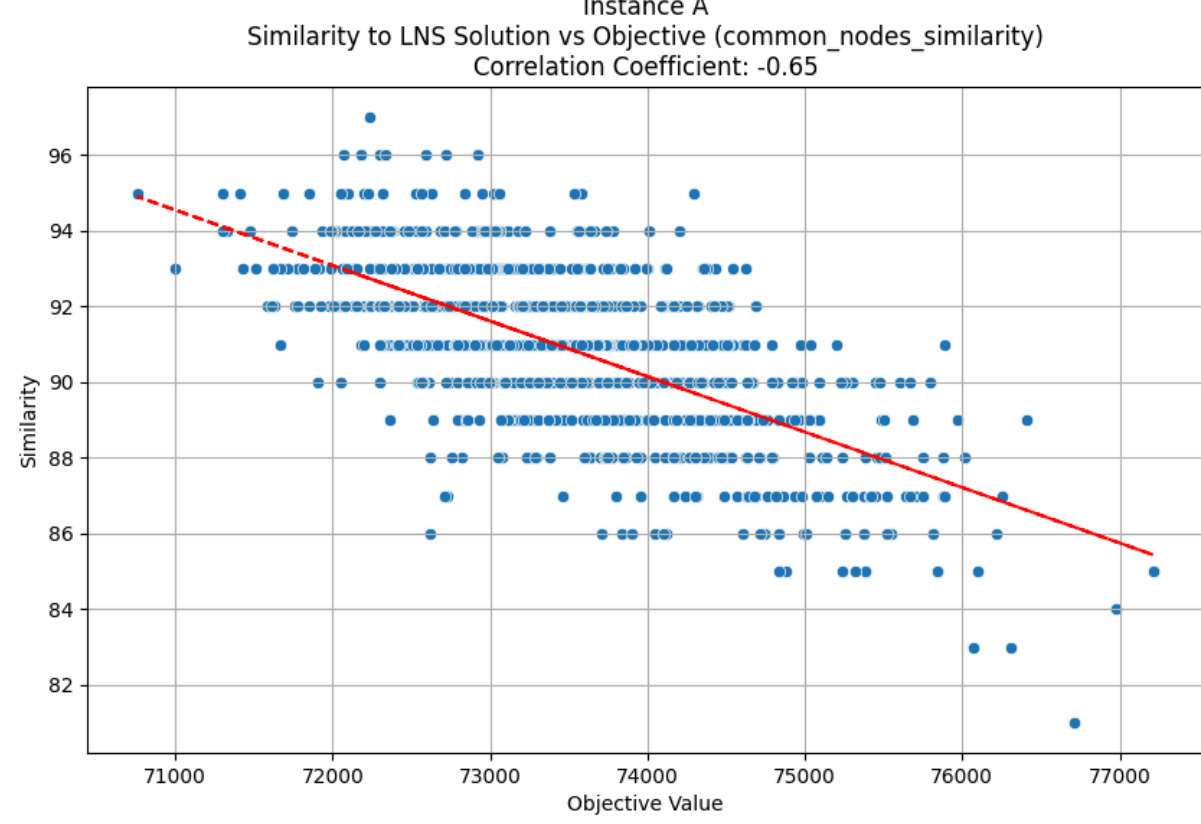
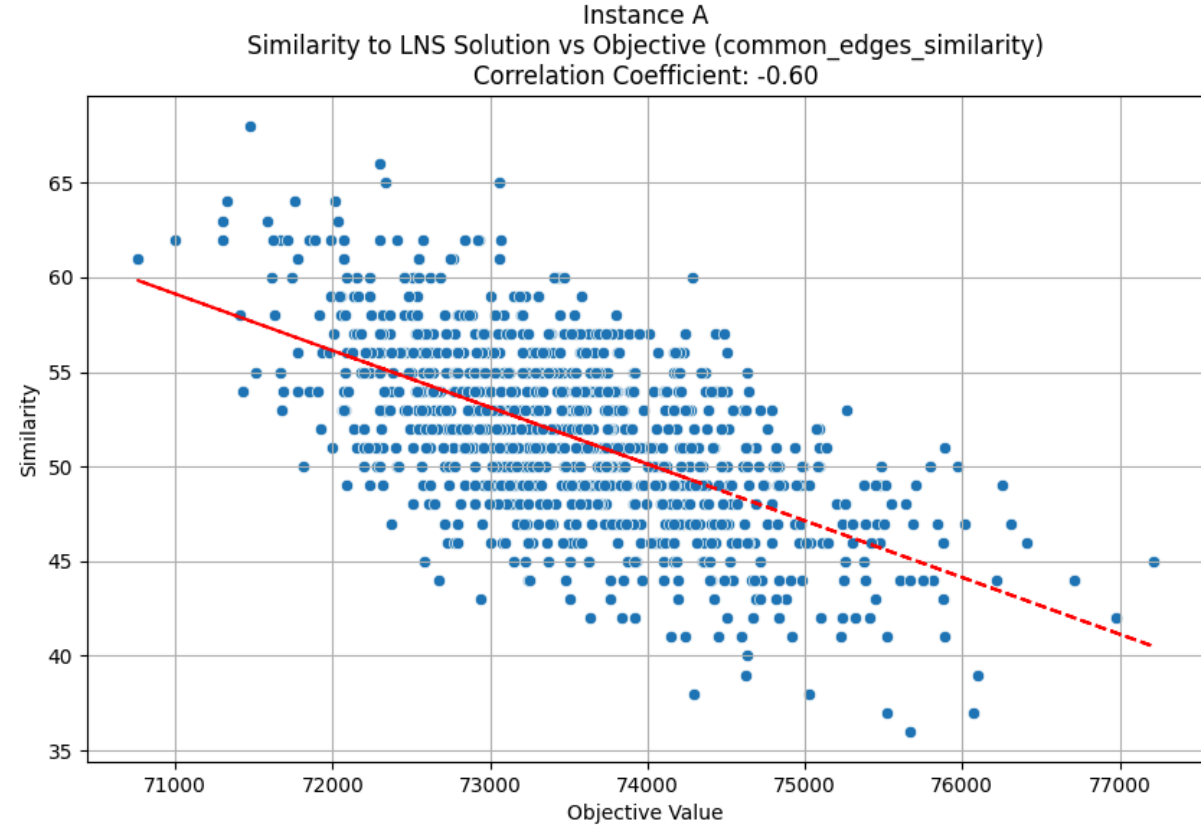
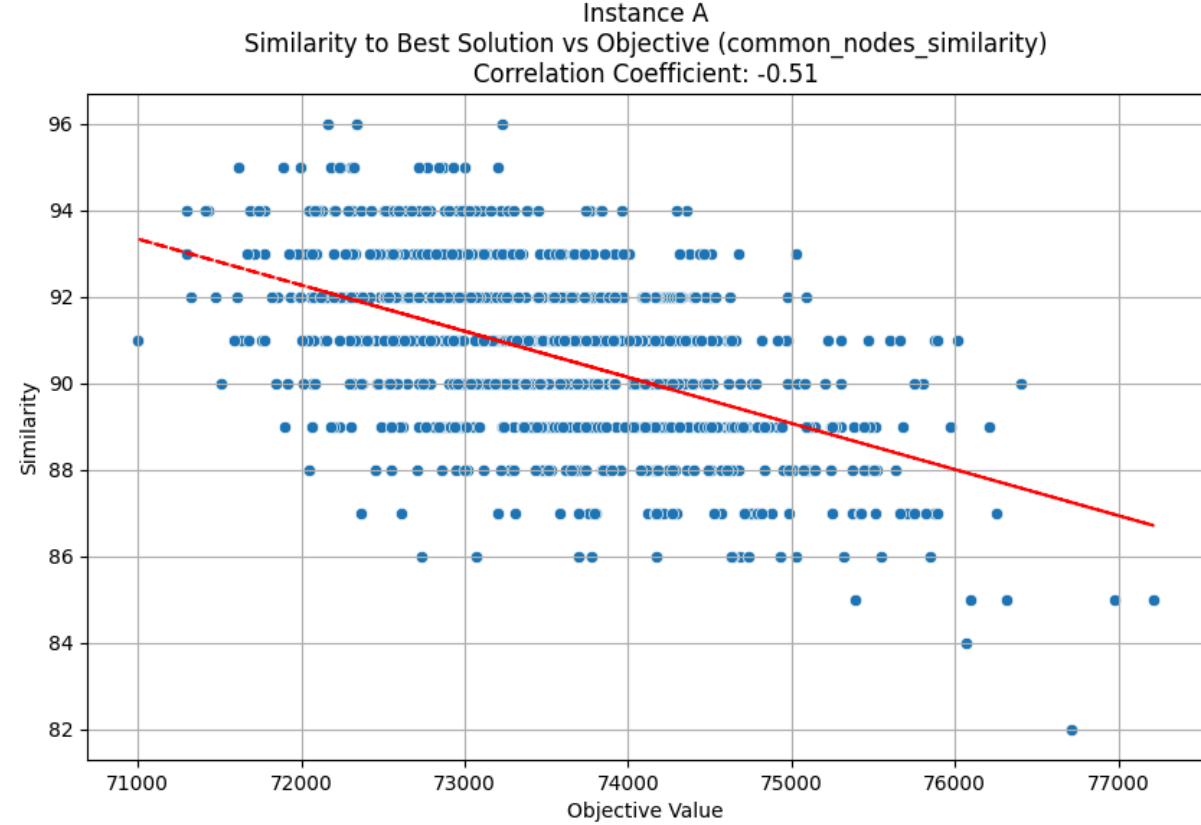
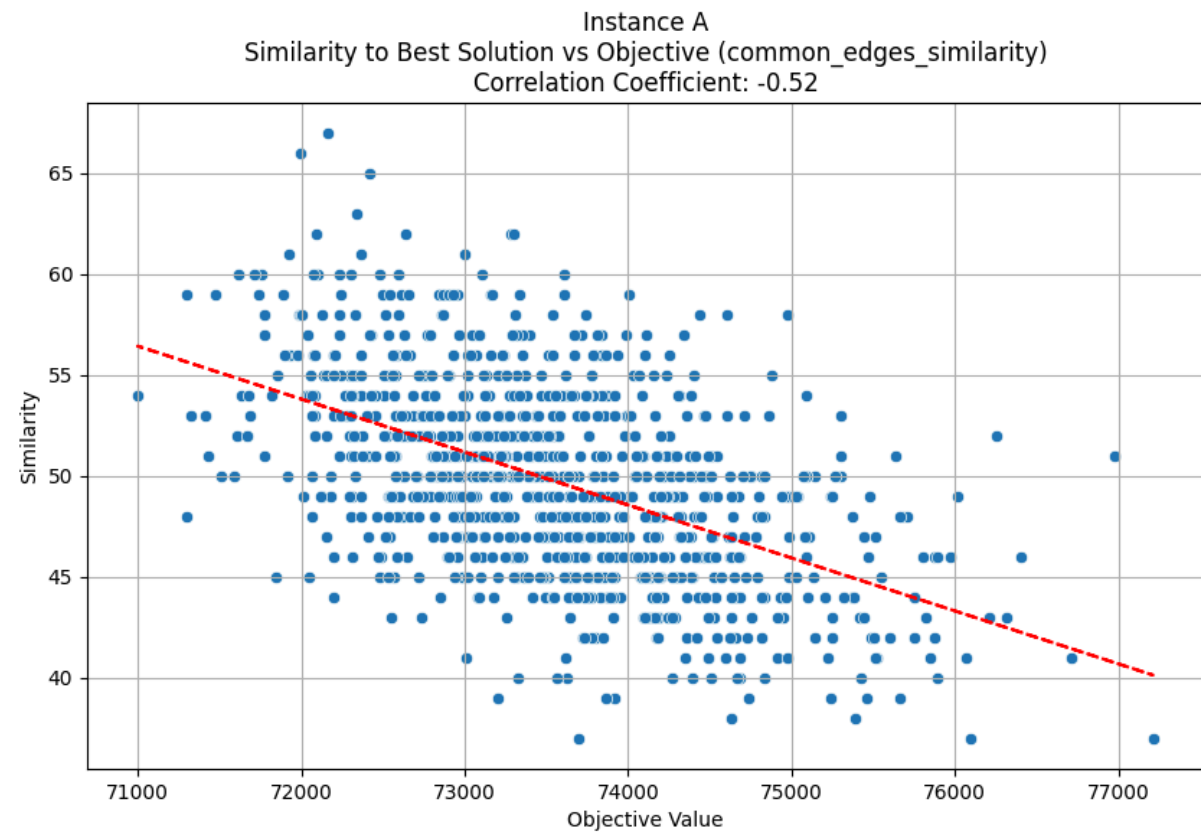
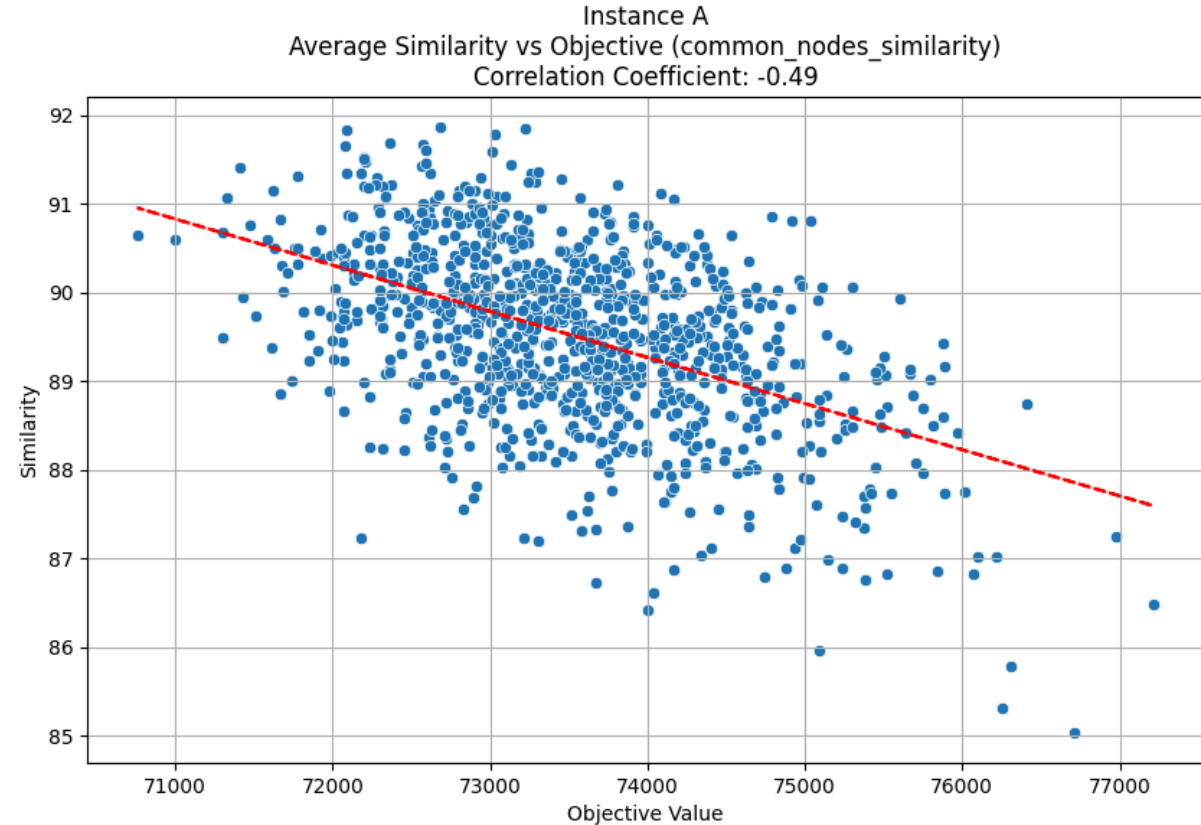
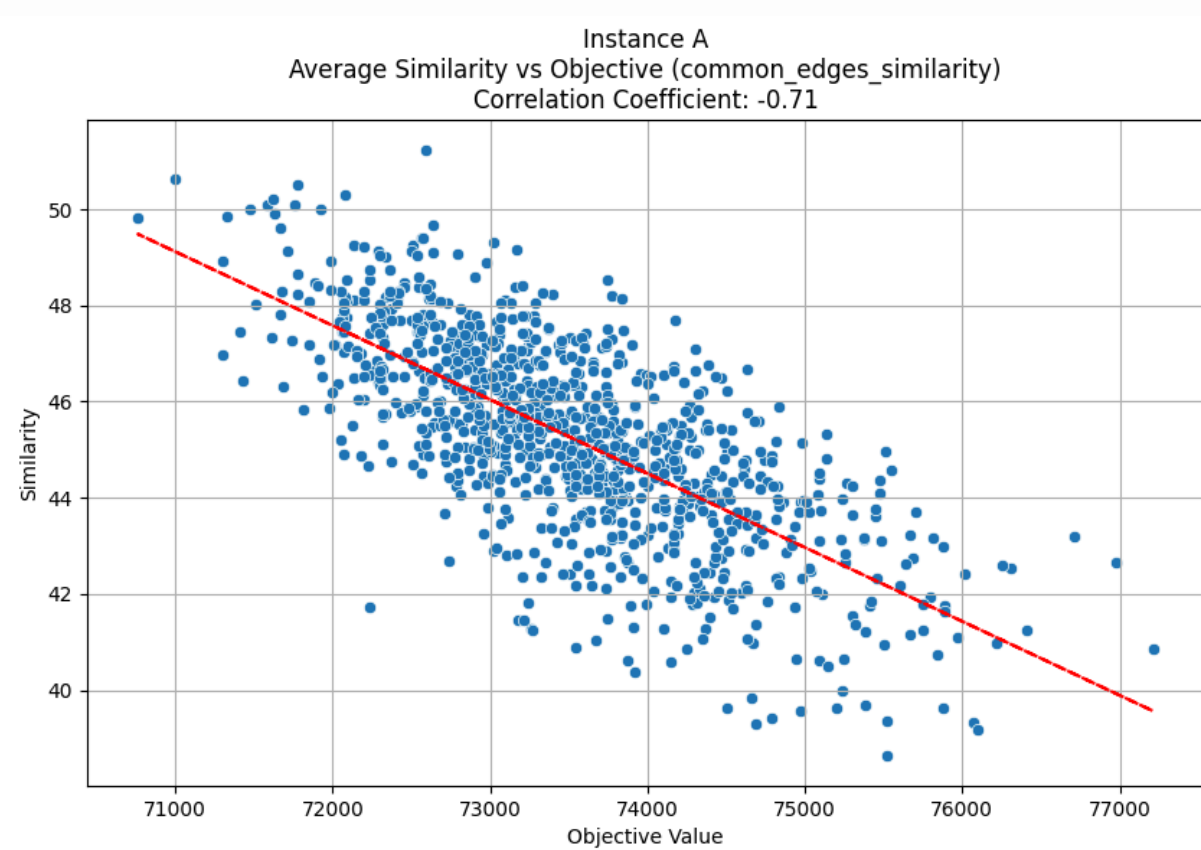
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Problem Description

The problem considered in this project involves a set of nodes, each represented by three integer values: two coordinates $(x,y)(x,y)$ that define the node's position in a two-dimensional plane, and a cost value associated with the node. The objective is to select exactly 50% of all available nodes (if the total number of nodes is odd, the number of selected nodes is rounded up) and construct a Hamiltonian cycle — that is, a closed path visiting each selected node exactly once and returning to the starting node. The goal is to minimize the sum of two components: 1. The total length of the constructed cycle, and 2. The total cost of all selected nodes. The distances between nodes are calculated using the Euclidean metric, rounded to the nearest integer.

Instance A



Instance B

